

阮劭伦 Shaolun RUAN

德国康斯坦茨 Abengbergweg 3 | ✉ slruan.2021@phdcs.smu.edu.sg
(+49) 0160 4464643 | 🌐 shaolun-ruan.com

研究简介 Research Profile

我致力于提升人们阅读与理解大规模数据的能力，通过设计新型图形表征与可视分析系统，帮助用户更高效、顺畅地借助机器开展数据分析。研究以**数据可视化 (Data Visualization)**和**人机交互 (Human-Computer Interaction)**为基础，聚焦面向人的计算工具，以解决不同科学领域中的复杂问题，促进模型可解释性与数据驱动决策。相关工具与设计已应用于医生、病理学家和量子计算开发者的实际工作。

教育与科研经历 Experience

University of Konstanz

博士后研究员

计算机与信息科学系；合作导师：*Daniel A. Keim* 教授；数据分析与可视化研究组成员

Konstanz, Germany

2026.05 – 至今

Monash University

研究学者

信息技术学院；合作导师：*Tim Dwyer* 教授；*Immersive Analytics Lab* 成员

Melbourne, Australia

2025.01 – 2025.12

Singapore Management University

博士

计算机与信息系统学院；导师：*Jiannan Li* 教授、*Yong Wang* 教授

Singapore

2022.01 – 2026.01

University of Electronic Science and Technology of China

学士

计算机科学与工程学院；大数据研究中心成员

Chengdu, China

2015.09 – 2019.07

荣誉与奖励 Honors and Awards

2025: Google 博士生奖学金 (Google PhD Fellowship)

研究方向：Advancing Explainable Quantum Computing through Visualization

2025: IEEE VIS 2025 最佳论文荣誉提名奖 (IEEE VIS 2025 Best Paper Honorable Mention)

获奖论文：“Qualitative Study for LLM-assisted Design Study Process: Strategies, Challenges, and Roles”

2024: 院长嘉奖名单 (SCIS Dean's List Award)

授予科研成果突出的前 3% 博士生

2024: SMU 校长奖学金 (SMU Presidential Doctoral Fellowship)

授予持续取得卓越科研成果的前 10% 博士生

2023: SMU 校长奖学金 (SMU Presidential Doctoral Fellowship)

授予持续取得卓越科研成果的前 10% 博士生

2019: UESTC 计算机科学与工程学院卓越毕业生奖 (UESTC SCSE Outstanding Student Award)

学生指导与学术论文 Mentorship and Publications

— Mentored Work —

Qipeng Wang(mentee), **Shaolun Ruan**(mentor), Rui Sheng, Yong Wang, Min Zhu, Huamin Qu.

TrajLens: Visual Analytics for Constructing Cell Development Trajectories in Cross-Sample Analysis.

IEEE Transactions on Visualization & Computer Graphics (2025).

— Mentored Work —

Qipeng Wang(mentee), Rui Sheng, **Shaolun Ruan**(mentor), Xiaofu Jin, Chuhan Shi, Min Zhu.

SynthLens: Visual Analytics for Facilitating Multi-step Synthetic Route Design.

IEEE Transactions on Visualization & Computer Graphics (2025).

— Mentored Work —

Shaolun Ruan(mentor), Rohan Ramakrishnar(mentee), Chao Ren, Rudai Yan, Qiang Guan, Jiannan Li, Yong Wang.
Towards Explainable Quantum AI: Informing the Encoder Selection of Quantum Neural Networks via Visualization.
IEEE Transactions on Visualization & Computer Graphics (2026).

— Mentored Work —

Shaolun Ruan(mentor), Ribo Yuan(mentee), Qiang Guan, Yanna Lin, Ying Mao, Weiwen Jiang, Zhepeng Wang, Wei Xu, Yong Wang.
Venus: A Geometrical Representation for Quantum State Visualization.
Computer Graphics Forum (CGF) 42(3). <https://doi.org/10.1111/cgf.14827>

Shaolun Ruan, Rui Sheng, Xiaolin Wen, Jiachen Wang, Tianyi Zhang, Yong Wang, Tim Dwyer, Jiannan Li.
Qualitative Study for LLM-assisted Design Study Process: Strategies, Challenges, and Roles.
IEEE Transactions on Visualization & Computer Graphics (2025).
IEEE VIS 2025 最佳论文荣誉提名奖 (Best Paper Honorable Mention)

Shaolun Ruan, Qiang Guan, Paul Griffin, Ying Mao, Yong Wang.
QuantumEyes: Towards Better Interpretability of Quantum Circuits.
IEEE Transactions on Visualization & Computer Graphics (2023): 1–13.
<https://doi.org/10.1109/TVCG.2023.3332999>

Shaolun Ruan, Zhiding Liang, Qiang Guan, Paul Griffin, Xiaolin Wen, Yanna Lin, Yong Wang.
Violet: Visual Analytics for Explainable Quantum Neural Networks.
IEEE Transactions on Visualization & Computer Graphics (2023).
<https://doi.org/10.1109/TVCG.2024.3388557>

Shaolun Ruan, Yong Wang, Weiwen Jiang, Ying Mao, Qiang Guan.
Vacsen: A Visualization Approach for Noise Awareness in Quantum Computing.
IEEE Transactions on Visualization & Computer Graphics 29.01 (2022): 462–472.
<https://doi.org/10.1109/TVCG.2022.3209455>

Shaolun Ruan, Yong Wang, Hailong Jiang, Weijia Xu, Qiang Guan.
BatchLens: A Visualization Approach for Analyzing Batch Jobs in Cloud Systems.
Proceedings of DATE 2022. IEEE, 2022: 108–111.
<https://ieeexplore.ieee.org/document/9774668>

Manusha Karunathilaka*, **Shaolun Ruan***, Lin-Ping Yuan, Jiannan Li, Zhiding Liang, Kavinda Athapaththu, Qiang Guan, Yong Wang.
Intuit: Explain Quantum Computing Concepts via AR-based Analogy.
ACM Conference on Human Factors in Computing Systems (Late-Breaking Work) (2025).
<https://dl.acm.org/doi/full/10.1145/3706599.3720085>

Shaolun Ruan, Yong Wang, Qiang Guan.
Intercept Graph: An Interactive Radial Visualization for Comparison of State Changes.
2021 IEEE Visualization Conference (VIS). IEEE, 2021: 111–115.
<https://doi.org/10.1109/VIS49827.2021.9623307>

Hailong Jiang*, **Shaolun Ruan***, Bo Fang, Yong Wang, Qiang Guan.
Visilience: An Interactive Visualization Framework for Resilience Analysis using Control-Flow Graph.
Proceedings of IEEE PRDC 2023. <https://ieeexplore.ieee.org/document/10356508>

Xiaolin Wen, Tai Nguyen, **Shaolun Ruan**, Qiaomu Shen, Jun Sun, Feida Zhu, Yong Wang.
PonxziLens+: Visualizing Bytecode Actions for Smart Ponzi Scheme Identification.
IEEE Transactions on Visualization & Computer Graphics (2024).
<https://ieeexplore.ieee.org/abstract/document/10794804>

Rui Sheng, Zelin Zang, Jiachen Wang, Yan Luo, Zixin Chen, Yan Zhou, **Shaolun Ruan**, Huamin Qu.
CellScout: Visual Analytics for Mining Biomarkers in Cell State Discovery.
IEEE Transactions on Visualization & Computer Graphics (2025).
<https://ieeexplore.ieee.org/abstract/document/11267149>

Can Liu, Wenjie Jiang, **Shaolun Ruan**, Kotaro Hara, Yong Wang.
Accessible Fine-grained Data Representation via Spatial Audio.
IEEE Computer Graphics and Applications (2026). <https://arxiv.org/abs/2604.08979>

Feng Liang, Xiaolin Wen, Shixian Zhou, **Shaolun Ruan**, Yuxuan Du, Qiang Guan, Jun Ye, Yong Wang.
Visualization for Quantum Computing: Current State and Future Directions.
Under review by IEEE TVCG. [ResearchGate]

Peng Kuai, Yukun Yang, **Shaolun Ruan**, Junchi Xu, Yanjie Zhang, Lin Zhang, Min Zhu, Rui Sheng.
Within the MDT Room: Situated in Multidisciplinary Team-Grounded Agent Debate for Clinical Diagnosis.
arXiv preprint: <https://arxiv.org/abs/2603.28393>

Shuo Yan, Xiaolin Wen, **Shaolun Ruan**, Yanjie Zhang, Jiaming Mi, Yushi Sun, Huamin Qu, Rui Sheng.
InconLens: Interactive Visual Diagnosis of Behavioral Inconsistencies in LLM-based Agentic Systems.
arXiv preprint: <https://arxiv.org/abs/2603.28106>

Shaolun Ruan, Zhikang Wang, Furui Cheng, Rui Sheng, Kai Qian, Jiangning Song, Jiannan Li, Yong Wang.
ImaGene: A Multimodal Visualization Tool for Gene-powered Cancer Prognosis.
Under review.

受邀报告 Invited Talks

- 2026.01: VIS Meets Quantum Computing**
ÉTS Quantum Seminar Series
- 2025.11: Visualization Towards Addressing the Explanation of Scientific Problems**
MIT University
- 2025.10: VIS Meets Quantum Computing**
University of Toronto
- 2025.06: Visualization Towards Addressing the Explanation of Scientific Problems**
University of Technology Sydney & Western Sydney University
- 2024.05: VIS Meets Quantum Computing**
GAMES Webinar
- 2024.04: VIS Meets Quantum Computing**
中南大学
- 2024.02: Stepping Into the Era of Interpretable Quantum Computing**
University of Notre Dame — QuCS Lecture Series
- 2024.01: Towards Making Your VIS Paper Writing Better**
电子科技大学
- 2024.01: VIS Meets Quantum Computing**
四川大学
- 2023.12: VAST Panel**
HKUST VisLab HAI Seminar
- 2023.11: VIS Meets Quantum Computing**
HKUST

教学经历 Teaching

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|--|-------------------|
| Studio 12, FIT3179 — Data Visualisation
<i>Faculty of Information Technology, Monash University</i>
Student Lecturer | 2025 Semester 2 |
| Studio 6, FIT3179 — Data Visualisation
<i>Faculty of Information Technology, Monash University</i>
Student Lecturer | 2025 Semester 2 |
| IS428 — Visual Analytics for Business Intelligence
<i>School of Computing and Information Systems, SMU</i>
Teaching Assistant | AY2023–24, Spring |
| CS711 — Reinforcement Learning
<i>School of Computing and Information Systems, SMU</i>
Teaching Assistant | AY2024–25, Fall |

媒体报道 Media

- [SMU News] Google Announces 2025 PhD Fellowship Recipients. [报道]
- [QuCS Lecture] No. 59, QuCS Lecture Series. [文章]
- [SMU SCIS News] SCIS PhD Dean’s List Award 2024. [报道]
- [SCU VIS News] Invited Talk at SCU VIS Lab. [中文报道]
- [GAMES Webinar] No. 324 — Talk and Panel Discussion. [中文报道]
- [UESTC News] The School of Computer Science Held the First “Outstanding Award” Ceremony. [中文报道]